

Siyuan Feng

✉ eric_feng2006@outlook.com



+86 185 2226 4229



Ericfeng0129

Education

University of Warwick

Sep 2024 – Present | BSc Mathematics

Tianjin Foreign Languages School Affiliated to Tianjin Foreign Studies University

Sep 2021 – Jun 2024

Internship Experience

Tianma Microelectronics Co., Ltd. Algorithm Engineering Intern Aug 2025 – Sep 2025

- Built a pixel-level E-paper video ghosting simulation pipeline in MATLAB and Python, covering video frame extraction, D/A drive mapping, and A/D grayscale output across **3** core modules; supported **1-bit** video input and **256-level** grayscale output.
- Designed a history-frame-difference-based hold-state driving algorithm to extend ghosting persistence through **3** drive states: positive, negative, and hold; replaced pixel-wise loops with NumPy vectorisation, reducing single-run video simulation time by approximately **85%**.

A-Level Mathematics & Cambridge STEP Tutor Apr 2024 – Aug 2024

- Delivered 100+ hours of tutoring; supported students in gaining offers from Oxford, Imperial College London, NUS, and other universities.

Projects & Research

Research Paper: A Symplectic-Form Proof of the Gottesman–Knill Theorem Jan 2026 – Mar 2026

- (*Published on Zenodo*) Conducted an independent study of the Gottesman–Knill theorem and classical simulation of quantum circuits; developed a symplectic linear-algebraic proof based on binary encodings and symplectic representations of Pauli operators.
- Paper link: [doi:10.5281/zenodo.19615972](https://doi.org/10.5281/zenodo.19615972)

Quantum Computing Project: Quantum Algorithm Modelling, Simulation and Variational Optimisation Dec 2025 – Jan 2026

- Built a quantum-circuit simulation environment using MindQuantum and completed the modelling, simulation, and basic validation of quantum algorithms.
- Reproduced representative quantum algorithms including QFT, QAOA, and VQE; implemented Max-Cut optimisation and ground-state energy estimation for the H₂ molecule.

Machine Learning Project: Implementation, Evaluation and Improvement of Regression and Classification Models Jan 2025 – Mar 2025

- Built regression and classification models on real-world datasets, conducting data cleaning, exploratory visualisation, baseline construction, and error diagnosis.
- Implemented linear regression, logistic regression, and a one-hidden-layer neural network using NumPy; compared the effects of different training settings on performance and generalisation.

Awards & Activities

- UK Undergraduate Calculus Competition: ranked 1st at the University of Warwick and represented the university at the national final.
- Bilibili calculus education content creator: 10,000+ followers.
- 4×4×4 Rubik's Cube: national ranking as high as Top 400.

Skills

- **Programming:** Python (NumPy / pandas / matplotlib / scikit-learn), MATLAB, R, Java
- **Tools & Environments:** Linux, Git (Gitee / AtomGit), Jupyter, Conda/Anaconda, VS Code
- **Command Line:** Windows PowerShell / CMD
- **Computer Vision:** OpenCV
- **Writing & Typesetting:** $\text{\LaTeX}$
- **Languages:** Chinese (Native), English (IELTS 7.5)